साद मुखर्जी अंतरराष्ट्रीय सूचना प्रौद्योगिकी संस्थान
Dr. Shyama Prasad Mukherjee International Institute of Information Technology, Naya Raipur
(Initiative of NTPC & Govt. of Chhattisgarh)
Email: info@iiitnr.ac.in, Tel: (0771) 2474040, Web: www.iiitnr.ac.in Est.2015

MID TERM Examination (Spring, 2026)

Branch & Year: B. Tech CSE **Semester: 4th**

Subject: Computer Architecture, **Sub Code: EC206C,** **Instructor Code: BCM**

Max Marks: 30 **Duration: 1.30 Hours** **Date: 24/02/2026**

Instruction: Answer **ANY THREE** the questions and to the point. All the parts of a question in same place.

1. (a) Why RISC philosophy is better? How ARM instruction set differs from the pure RISC in several ways? [5]

(b) Write the difference between RISC and CISC architecture? Why Harvard architechture is better than Von Neumann architecture? [5]

2. (a) Write one ARM instruction set of addition and multiplication. What is the role of barrel shifter? [5]

(b) Why jazzle processing is there in ARM processing? Explain the Amdahl's Law with an example. [5]

3. (a) Write short note on processor mode, current program status register, and thump instruction. [6]

(b) What are the hardware fundamentals of ARM processor? [4]

4. (a) Why ARM processors are not found in leading-edge high performance? [4]

(b) Why pipelining is important? Give one example of ARM 9 pipelining with diagram. [6]

Best of Luck

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